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Тенденции развития искусственного интеллекта в области анализа данных с использованием машинного и глубокого обучения

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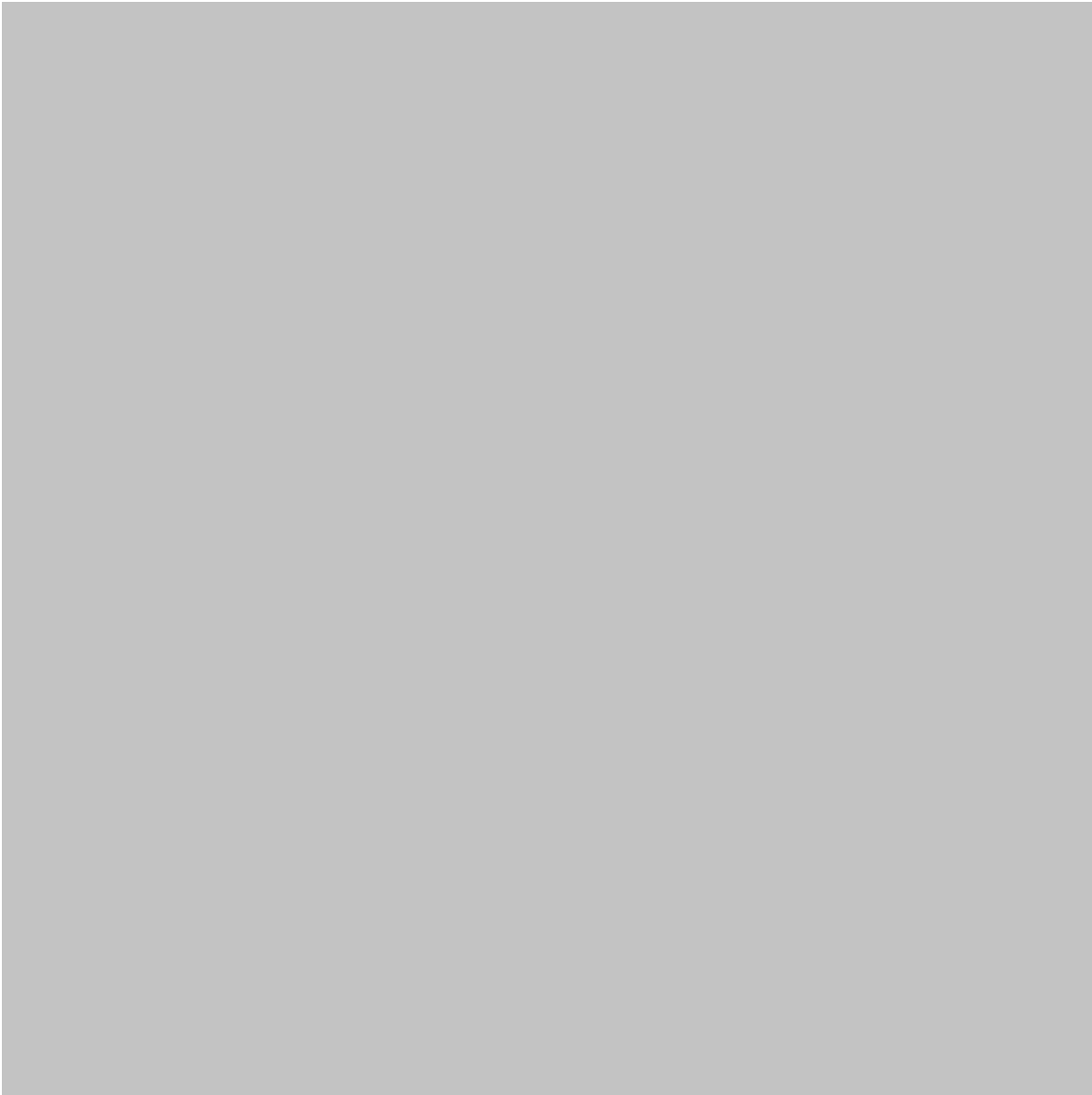
Тенденции развития искусственного интеллекта в области анализа данных с использованием машинного и глубокого обучения

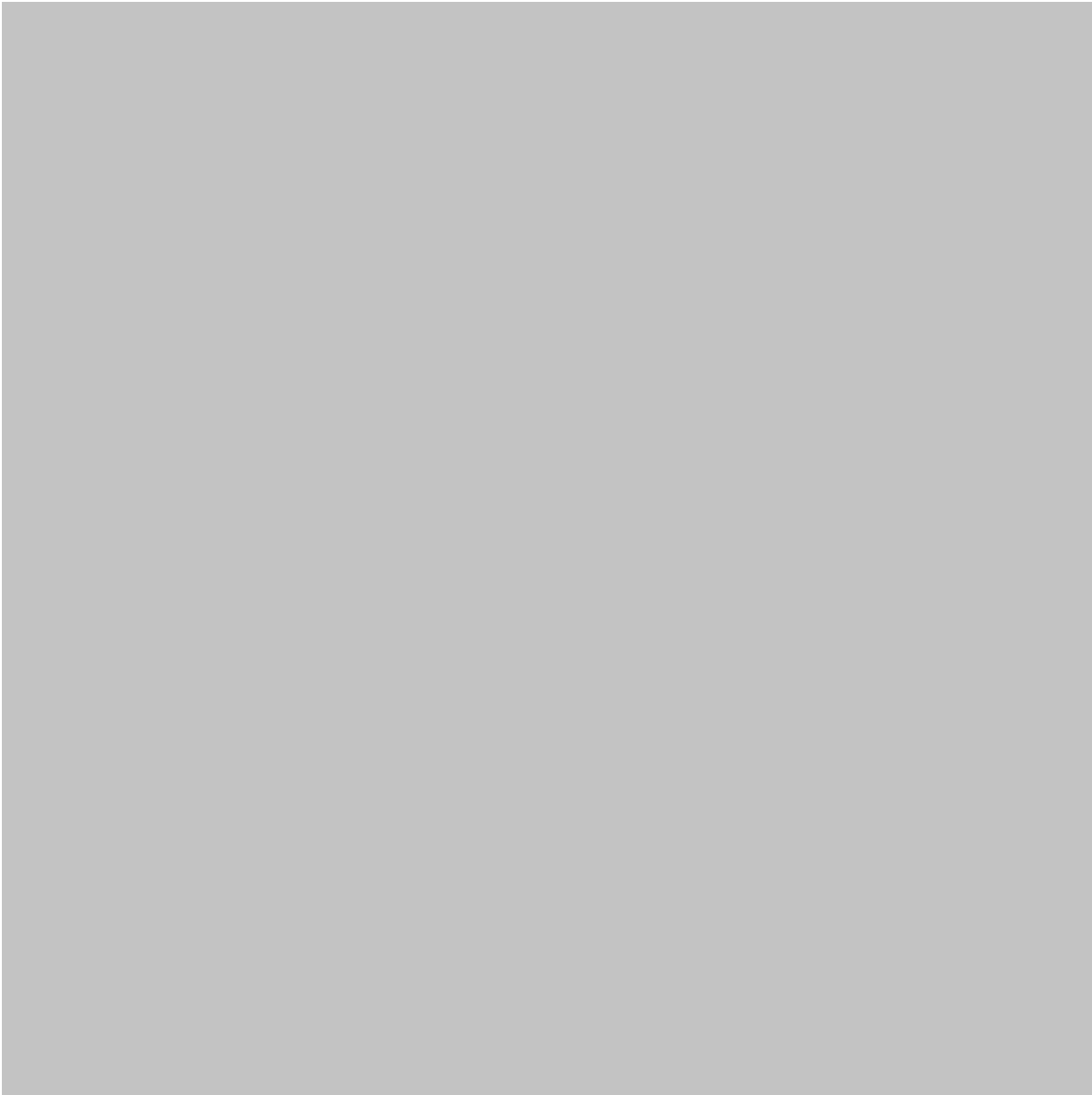
Введение

Аналитика данных переходит от дисциплины, ориентированной на составление отчетов, к интеллектуальной системе принятия решений. Традиционная аналитика в основном отвечает на такие вопросы, как *Что произошло?* и *Почему это произошло?* Современный искусственный интеллект (ИИ), машинное обучение (МО) и глубокое обучение (ГО) все чаще помогают организациям находить ответы на более сложные вопросы: *Что, скорее всего, произойдет дальше?* *Какие действия нам следует предпринять?* *Может ли система автоматически выявить закономерность?*

Эта трансформация особенно важна для организаций, работающих в сфере инженерии, финансов, производства, здравоохранения, логистики, энергетики, розничной торговли, телекоммуникаций и технологий, где постоянно генерируются большие объемы структурированных и неструктурированных данных.

Недавние отраслевые исследования показывают, как быстро меняется ситуация. Согласно индексу развития искусственного интеллекта Стэнфордского университета за 2026 год, возможности ИИ стремительно расширяются, а его внедрение становится все более распространенным в организациях. По мнению компании Garner, основными тенденциями в области данных и аналитики в 2026 году станут ИИ-агенты, семантические технологии и конвергентные платформы для работы с данными и аналитикой.





Важный инженерный принцип прост:

Более совершенный искусственный интеллект не означает автоматически более совершенную аналитику.

Высококачественные данные, подходящие модели, надежная инфраструктура, управление, знание предметной области и проверка человеком по-прежнему играют важную роль. В исследовании компании Garner за 2026 год особо подчеркивается, что традиционные методы прогнозирования, такие как экстраполяция и классификация, по-прежнему приносят значительную практическую пользу в сфере ИИ.

Теоретическая база

От описательной аналитики к интеллектуальной

Аналитика данных традиционно развивалась в несколько этапов:

- **Описательная аналитика** → Что произошло?
- **Диагностическая аналитика** → Почему это произошло?
- **Предиктивная аналитика** → Что может произойти?
- **Предписывающая аналитика** → Что нам следует делать?
- **Автономная аналитика** → Может ли система постоянно давать рекомендации или принимать решения?

Машинное обучение особенно полезно на этапе прогнозирования, поскольку алгоритмы могут выявлять закономерности на основе исторических наблюдений, а не полагаться исключительно на правила, написанные вручную.

Глубокое обучение расширяет эти возможности за счет многослойных нейронных сетей, которые могут формировать сложные представления на основе больших массивов данных. Современные системы искусственного интеллекта также могут обрабатывать комбинации текста, изображений, аудио, показаний датчиков и обычных табличных данных.

Взаимосвязь между искусственным интеллектом, машинным обучением и глубоким обучением

Полезный способ понять иерархию:

Искусственный интеллект

↳ Машинное обучение

↳ Глубокое обучение

↳ Нейронные сети / специализированные архитектуры

Таким образом, машинное обучение неотделимо от искусственного интеллекта и является одним из основных подходов к созданию интеллектуальных систем.

Глубокое обучение — это специализированный подход к машинному обучению, основанный на многослойных нейронных сетях.

Для аналитиков данных это различие имеет значение, поскольку не для каждой аналитической задачи требуется глубокое обучение. Иногда относительно простая модель может превзойти сложную нейронную сеть, если набор данных небольшой, структурированный или легко интерпретируемый.

Определение

Что такое аналитика данных на основе искусственного интеллекта?

Аналитика данных на основе искусственного интеллекта — это применение методов искусственного интеллекта для сбора, преобразования, понимания, прогнозирования и передачи информации, содержащейся в данных.

Она объединяет в себе несколько технологий:

- Машинное обучение
- Глубокое обучение
- Обработка естественного языка
- Компьютерное зрение
- Генеративный ИИ
- Автоматизированное машинное обучение
- Обнаружение аномалий
- Предиктивное прогнозирование
- Рекомендательные системы
- Агенты ИИ
- Графы знаний и семантические технологии

Цель состоит не просто в автоматизации расчетов. Цель состоит в том, чтобы извлекать полезную информацию из все более сложных наборов данных и преобразовывать ее в практические решения.

Машинное обучение в аналитике

Алгоритмы машинного обучения могут выявлять взаимосвязи в исторических данных и использовать их для классификации наблюдений, прогнозирования будущих событий, обнаружения аномалий или выработки рекомендаций.

К типичным областям применения относятся:

- Прогнозирование оттока клиентов
- Прогнозирование спроса
- Выявление мошенничества
- Прогнозирование выхода оборудования из строя
- Анализ кредитного риска
- Прогнозирование энергопотребления
- Мониторинг контроля качества

Глубокое обучение в аналитике

Глубокое обучение становится особенно ценным, когда данные содержат сложные закономерности.

Например, производственная компания может анализировать:

- Вибрация машины
- Тепловизионные изображения
- Акустические сигналы
- Видеопотоки
- Временные ряды с датчиков

A deep neural network can potentially discover useful features directly from these inputs instead of requiring engineers to manually define every feature.

Step-by-Step Explanation: Building an AI Analytics System

Step 1: Define the business or engineering problem

Start with the decision rather than the algorithm.

Instead of saying:

“We need deep learning.”

Define the objective:

“We need to identify machines likely to fail before production is interrupted.”

This difference prevents unnecessary technological complexity.

Step 2: Collect relevant data

Potential data sources include:

- Databases
- IoT sensors
- APIs
- Transaction systems
- Enterprise applications
- Images
- Documents
- Customer interactions
- Web activity
- Industrial equipment

The data should be relevant to the target problem.

Step 3: Clean and prepare the data

Raw data frequently contains:

- Missing values
- Duplicate records
- Incorrect measurements
- Outliers
- Inconsistent formats
- Incorrect labels
- Sensor errors

Data preparation can have a larger effect on the final system than switching between similar algorithms.

Step 4: Explore the dataset

Exploratory data analysis helps engineers understand:

- Distributions
- Trends
- Correlations
- Seasonality
- Anomalies
- Data imbalance
- Relationships between variables

Visualization remains important even when sophisticated AI models are used.

Step 5: Select an appropriate model

Possible choices include:

Traditional ML

- Linear models
- Decision trees
- Random forests
- Gradient boosting
- Support vector machines
- Clustering algorithms

Deep learning

- Convolutional neural networks
- Recurrent neural networks
- Transformers
- Autoencoders
- Specialized time-series architectures

Modern AI

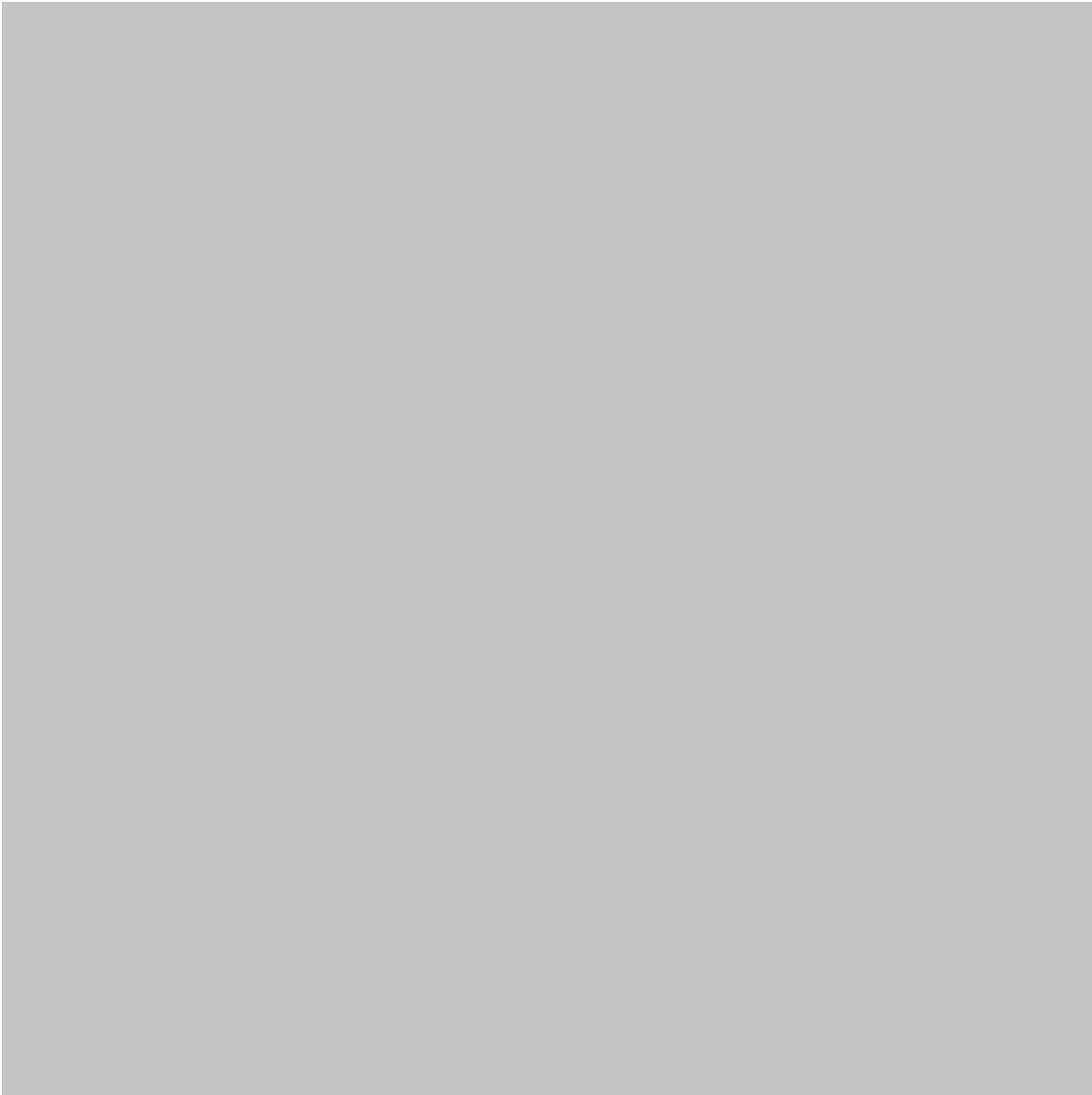
- Large language models
- Multimodal models
- AI agents
- Retrieval-augmented systems

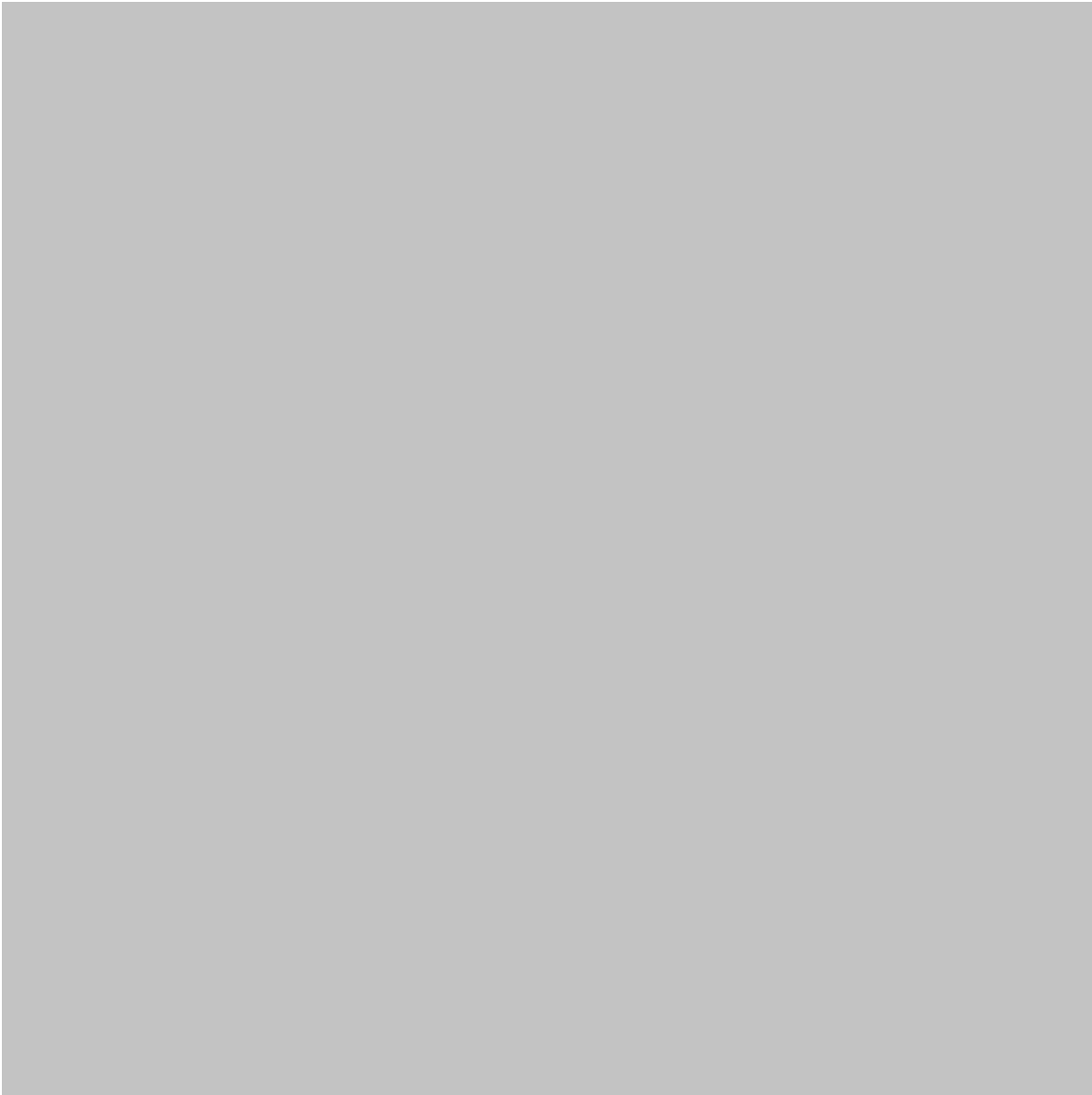
The best model is not necessarily the most complicated one.

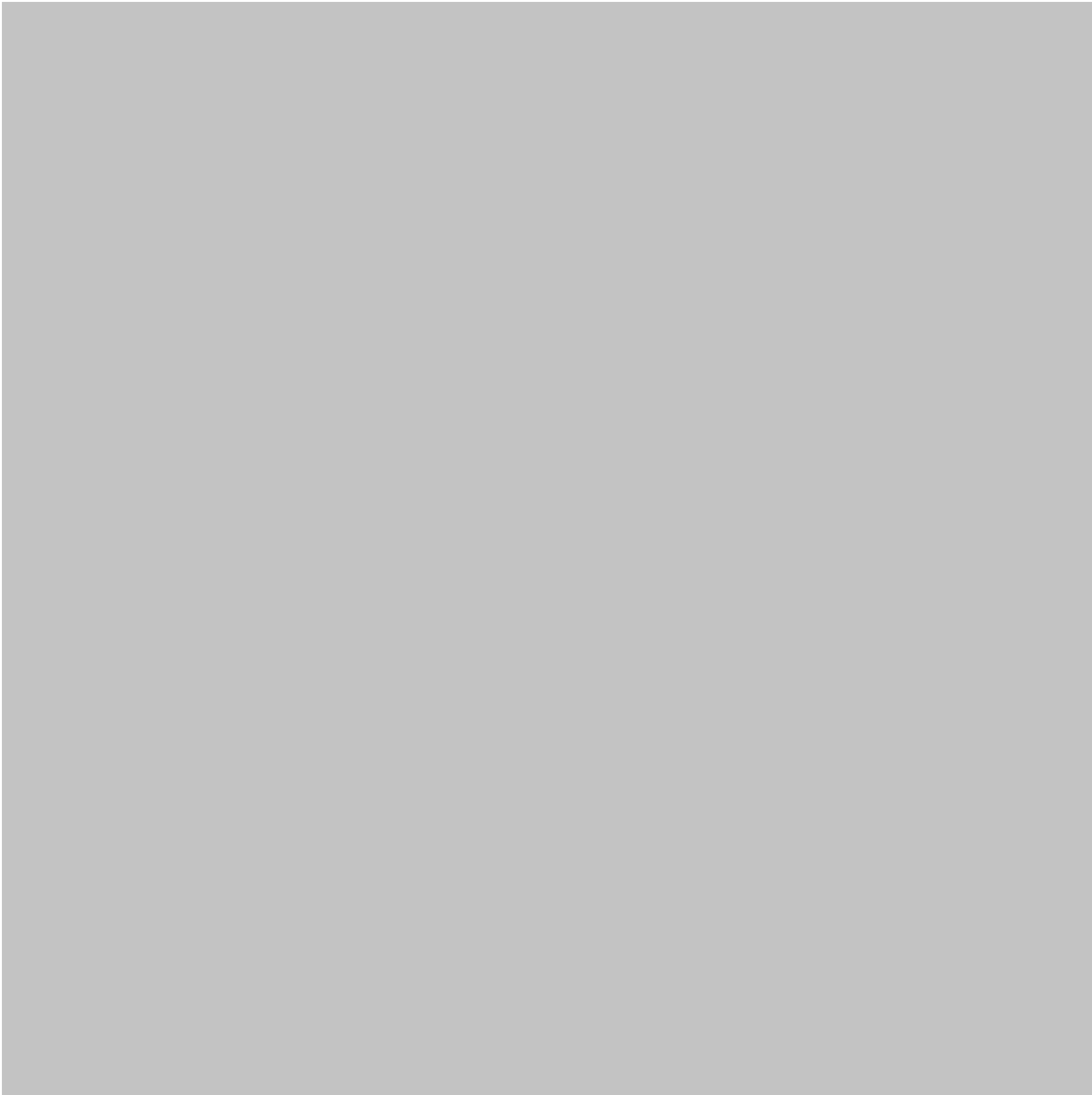
Step 6: Train and validate

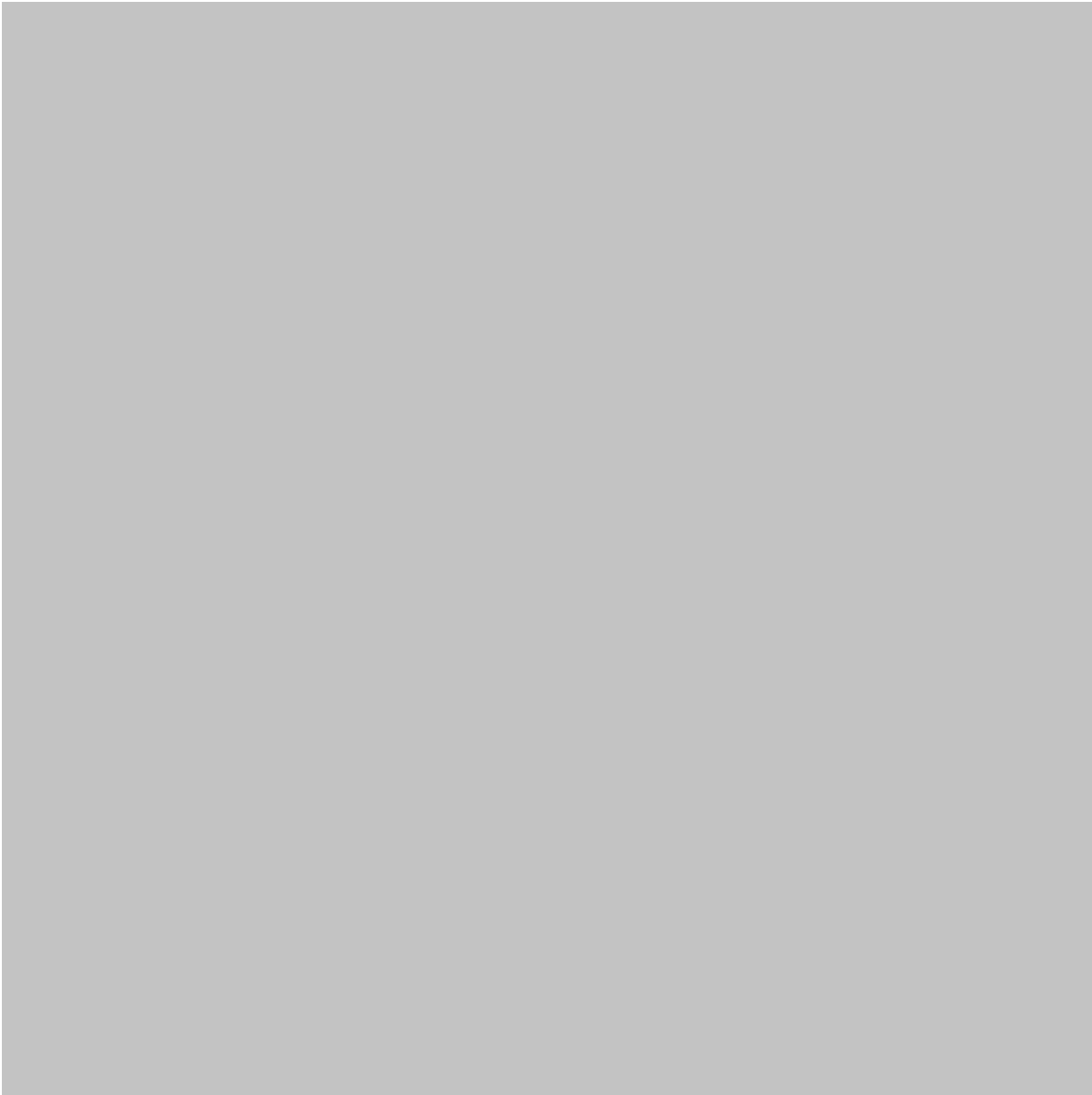
The dataset should be separated into appropriate training and validation/testing portions.

Engineers should evaluate whether the model generalizes to new data instead of simply memorizing historical observations.









Step 7: Deploy the analytics system

A model becomes useful when it can operate inside a real workflow.

Deployment may involve:

Data → Processing → Model → Prediction → Dashboard → Human/Automated Decision

For example, an industrial analytics platform might continuously ingest sensor readings, detect abnormal behavior, generate a risk alert, and notify a maintenance team.

Step 8: Monitor the model

Deployment is not the end.

Real-world data changes.

Customer behavior changes. Equipment ages. Markets move. Sensors are replaced. Regulations evolve.

Therefore, engineers need continuous monitoring for:

- Model drift
- Data drift
- Prediction quality
- Latency
- Bias
- Security
- Cost
- Reliability

Major AI Trends in Data Analytics

AI agents for analytics

One of the most important emerging trends is the use of AI agents that can perform multiple analytical tasks.

An analytics agent could potentially:

1. Receive a question.
2. Identify relevant datasets.
3. Query a database.
4. Analyze the results.
5. Generate visualizations.
6. Explain anomalies.
7. Produce a report.
8. Recommend next actions.

Garner identifies AI agents as a central 2026 data-and-analytics trend, while emphasizing the importance of decision governance as agents take on more operational responsibilities.

Natural-language analytics

Traditional analytics requires users to understand SQL, dashboard interfaces, or specialized statistical tools.

Natural-language analytics changes the interaction model.

A manager might ask:

“Which product categories experienced the largest decline in European sales last quarter?”

An AI analytics system could translate that request into data operations and return a visual explanation.

However, natural-language interfaces should not eliminate validation. A fluent answer can still be technically incorrect.

Predictive analytics remains essential

Despite the popularity of generative AI, conventional machine learning remains highly valuable.

Forecasting, classification, anomaly detection, and optimization are often directly connected to measurable business outcomes. Garner’s 2026 machine-learning research specifically notes that forecasting and classification currently deliver substantial AI value.

Multimodal analytics

Organizations increasingly have multiple forms of data.

Consider a factory:

Sensor data + maintenance records + images + video + technician notes

Instead of analyzing each source independently, multimodal AI can potentially combine them.

This creates richer analytical context.

Synthetic data

Synthetic data is becoming an important technique for training and testing AI systems, especially when real datasets are limited or sensitive.

However, synthetic data should not automatically be treated as equivalent to real-world observations. Current debate highlights the risk that generated datasets may fail to represent unexpected real-world behavior.

Explainable and responsible AI

As AI becomes involved in important decisions, organizations increasingly need to understand:

- Why was this prediction generated?
- Which data influenced the result?
- Can the result be audited?
- Is the model biased?
- Who is responsible when it fails?

Responsible AI therefore becomes an engineering requirement rather than merely an ethical discussion.

Comparison: ML vs Deep Learning vs Generative AI

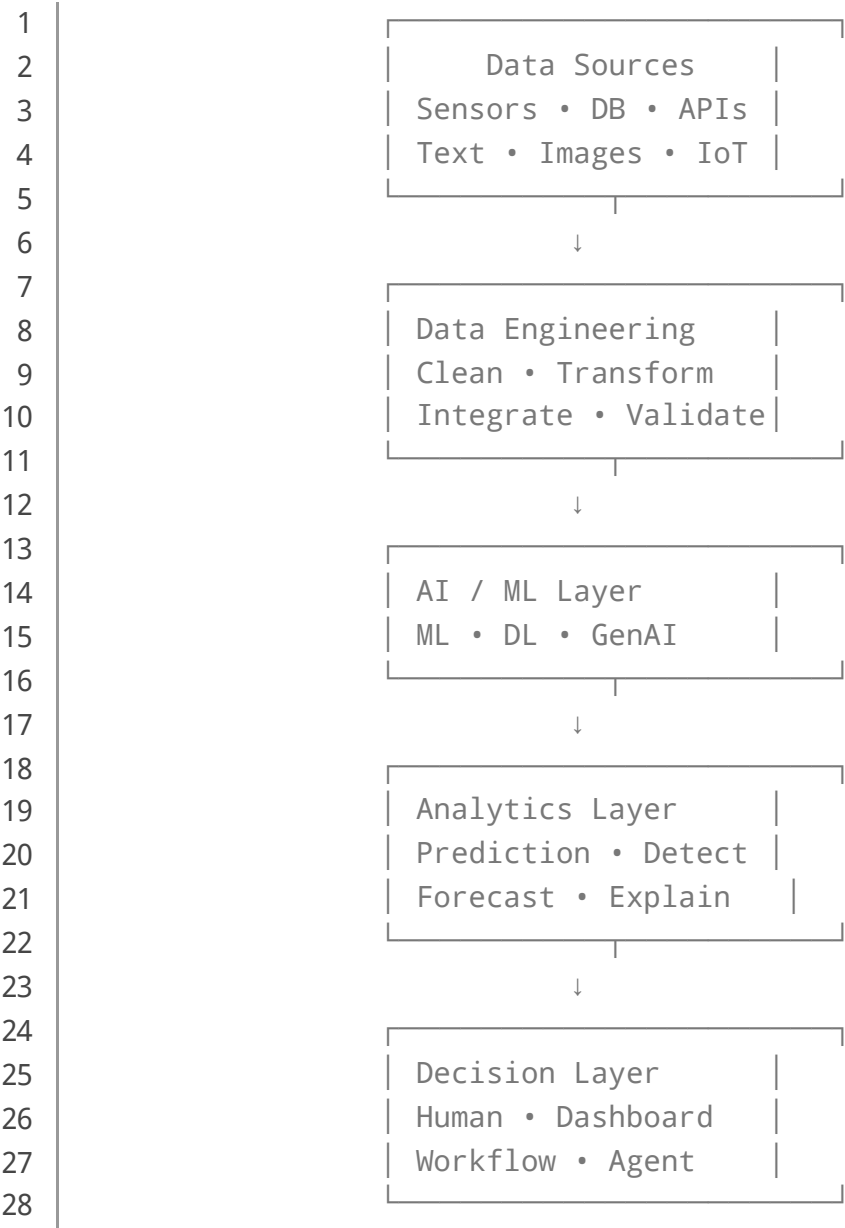
Feature	Machine Learning	Deep Learning	Generative AI
Typical data	Structured/tabular	Large complex datasets	Text, images, audio, multimodal
Data requirement	Low-medium	Usually medium-high	Often very high
Interpretability	Often higher	Often lower	Variable
Training complexity	Moderate	High	Very high
Common use	Forecasting, classification	Vision, speech, complex patterns	Content, reasoning, assistants
Hardware needs	Often modest	Often substantial	Frequently substantial
Best strength	Structured prediction	Representation learning	Generating and interacting with information

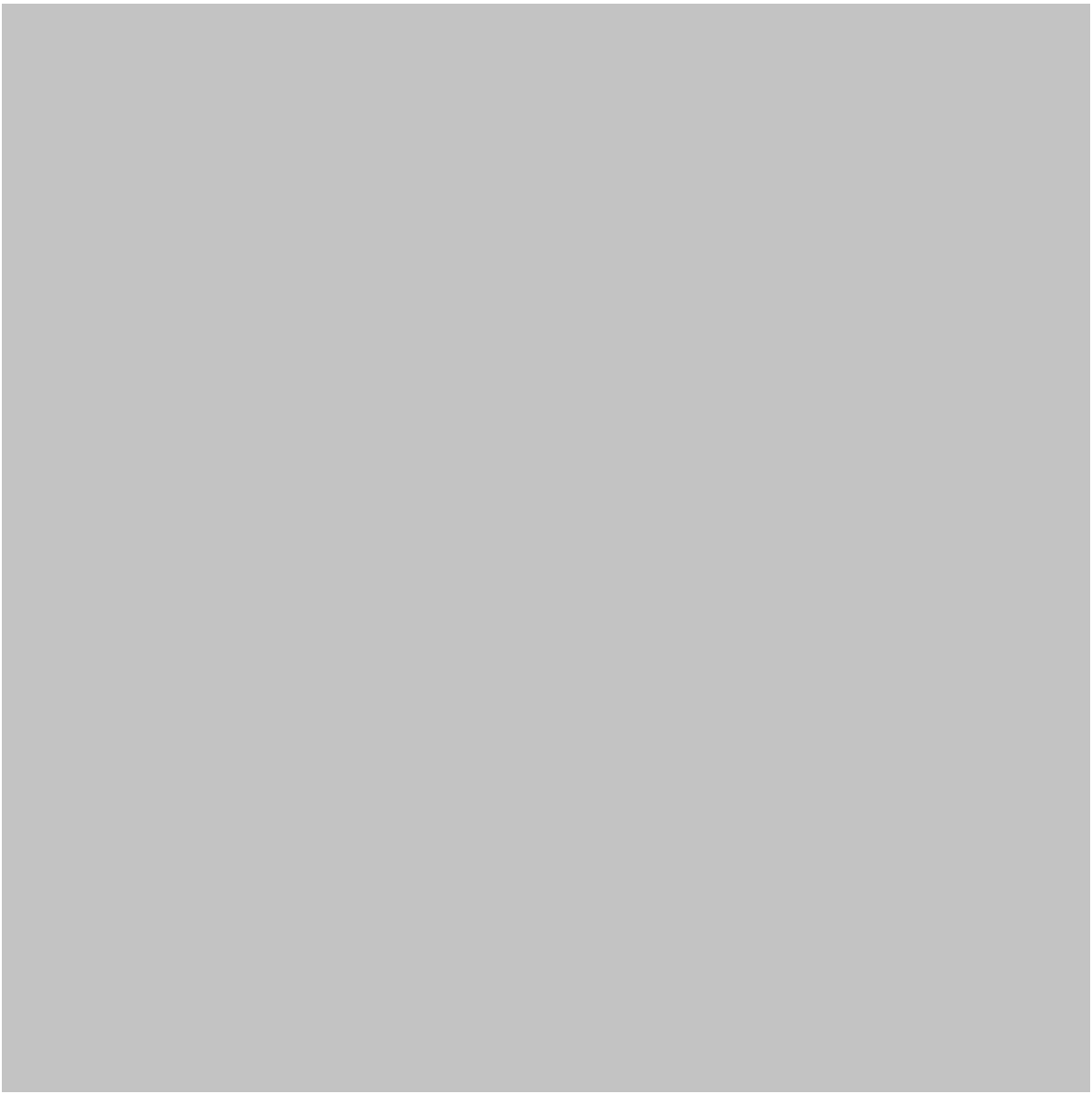
The practical lesson is important:

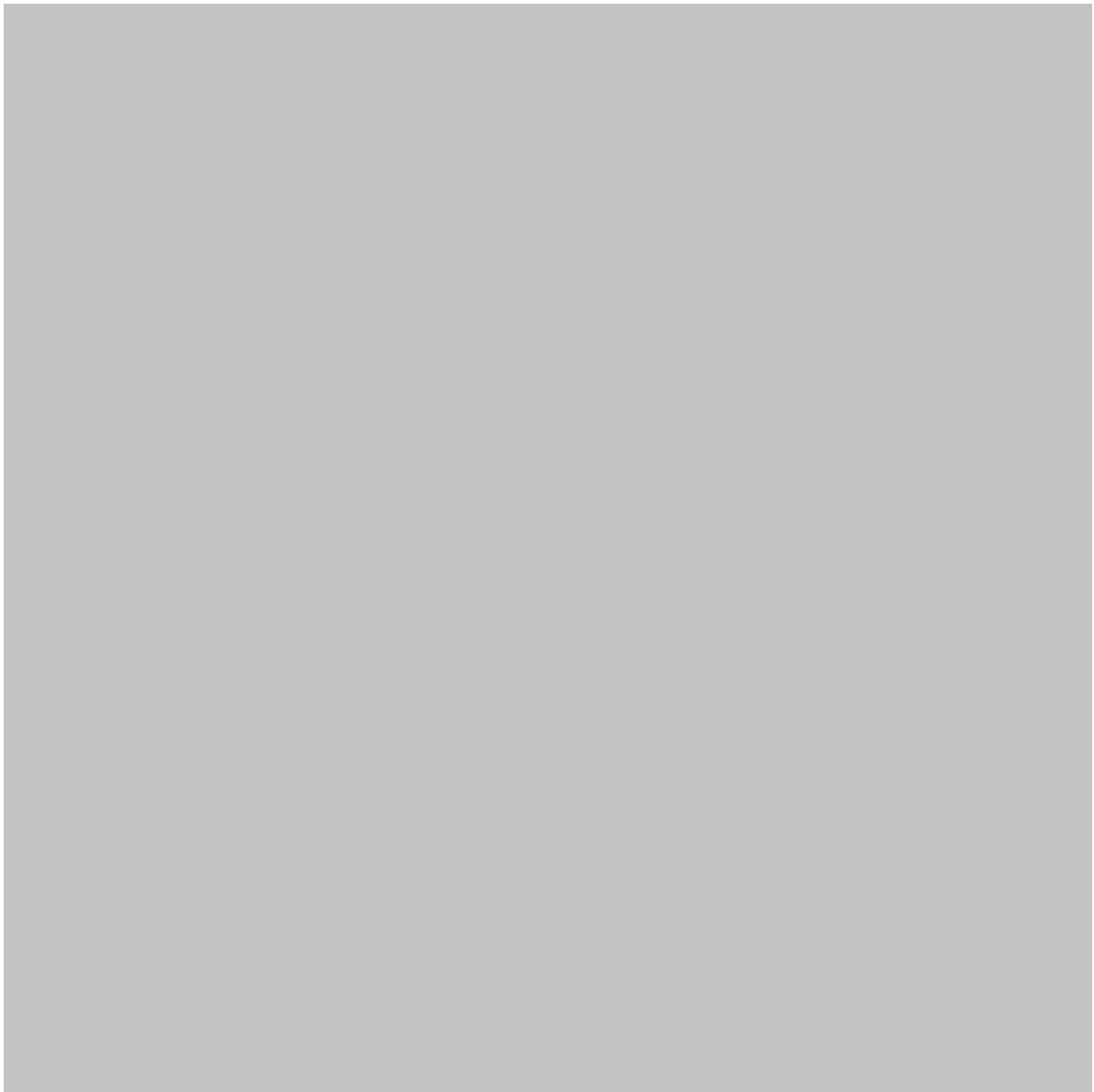
Choose the technology based on the problem—not the popularity of the technology.

Diagrams & Tables

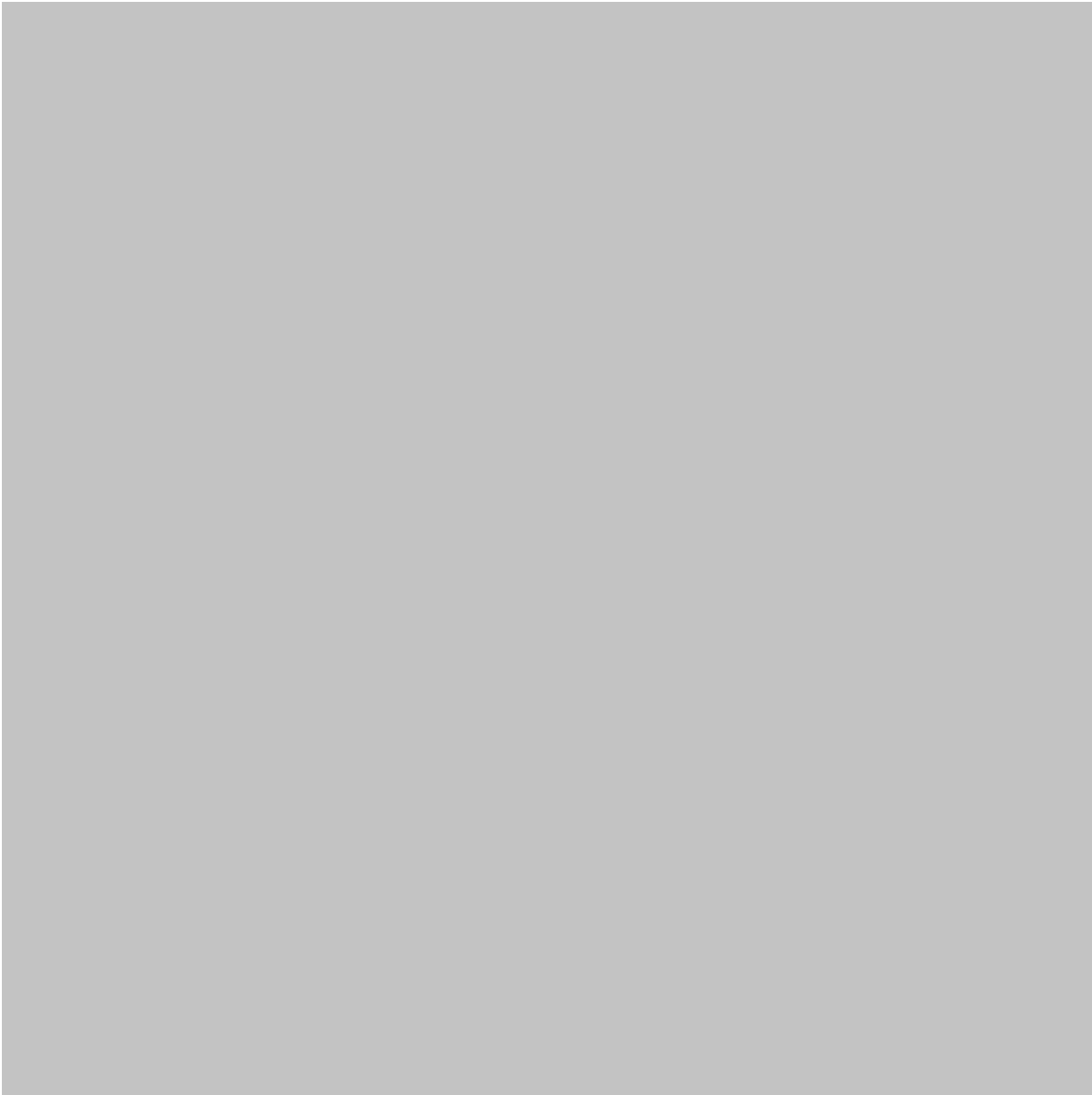
AI analytics architecture











Traditional analytics vs AI analytics

Capability	Traditional Analytics	AI-Enhanced Analytics
Reporting	Strong	Strong
Pattern discovery	Limited	Strong
Prediction	Statistical/manual	Automated ML/DL
Natural language	Limited	Advanced
Unstructured data	Limited	Strong
Automation	Moderate	High
Continuous learning	Usually limited	Possible

Capability	Traditional Analytics	AI-Enhanced Analytics
Autonomous decisions	Rare	Increasing

Examples

Example 1: Predictive maintenance

A wind-farm operator collects vibration, temperature, power-output, and maintenance data.

A machine-learning model learns patterns associated with equipment degradation.

Instead of waiting for a turbine to fail, the maintenance team receives an early warning.

The result can be:

Sensor data → AI prediction → maintenance alert → inspection → planned repair

Example 2: Retail demand analytics

A retailer combines historical sales, promotions, product categories, seasonal behavior, and inventory information.

An AI system can forecast demand and identify products that may experience unusual demand.

Managers can then adjust inventory before shortages occur.

Example 3: Engineering document analytics

An engineering organization may have thousands of reports, inspection documents, specifications, and maintenance records.

Natural-language AI can help engineers locate relevant information and summarize technical documents.

The system does not replace engineering judgment; it reduces the time required to locate and organize information.

Real-World Applications

Manufacturing

AI analytics can support:

- Predictive maintenance
- Defect detection
- Production optimization
- Energy monitoring
- Quality control

Healthcare

Potential applications include:

- Medical-image analysis
- Patient-risk prediction
- Hospital resource forecasting
- Drug discovery
- Clinical decision support

Healthcare systems require particularly strong validation, privacy, governance, and human oversight.

Finance

Financial organizations use AI analytics for:

- Fraud detection
- Risk assessment
- Customer segmentation
- Market analysis
- Transaction monitoring

Energy

AI can analyze:

- Electricity demand
- Renewable-energy production
- Equipment performance
- Grid conditions
- Consumption patterns

Transportation

Applications include:

- Fleet optimization
- Predictive maintenance
- Traffic forecasting
- Route optimization
- Logistics planning

The World Bank's 2026 World Development Report highlights practical AI applications including medical screening, weather forecasting for agriculture, and educational support.

Common Mistakes

Choosing deep learning too early

A small structured dataset may not require a neural network.

Start with a baseline model and increase complexity only when necessary.

Ignoring data quality

A sophisticated model trained on poor-quality data can produce unreliable predictions.

Garbage in → unreliable intelligence out.

Optimizing only for accuracy

Accuracy alone may not be sufficient.

An engineering system also needs to consider:

- Reliability
- Latency
- Cost
- Interpretability
- Safety
- Maintenance

Forgetting data drift

A model trained on yesterday’s environment may perform poorly tomorrow.

Continuous monitoring is essential.

Trusting generated explanations blindly

Generative AI can produce convincing explanations that are not necessarily correct.

Critical decisions require independent validation.

Challenges & Solutions

Challenge	Engineering Solution
Poor data quality	Build strong data-validation pipelines
Model drift	Continuous performance monitoring
High computing costs	Optimize models and infrastructure
Lack of explainability	Use interpretable models or explanation techniques
Privacy risks	Apply access controls and privacy-preserving methods
AI hallucinations	Ground outputs in verified data
Integration problems	Use APIs and standardized interfaces
Organizational resistance	Train users and demonstrate measurable outcomes

AI infrastructure is also becoming a major economic consideration. Garner forecasts worldwide AI spending at approximately \$2.59 trillion in 2026, with infrastructure representing a major share of that expenditure.

Case Study: AI-Based Predictive Maintenance

The engineering problem

Imagine a manufacturing facility operating hundreds of industrial machines.

Historically, maintenance is performed according to fixed schedules.

The problem is that some machines receive maintenance even when they are operating normally, while others may fail before their scheduled maintenance date.

The AI approach

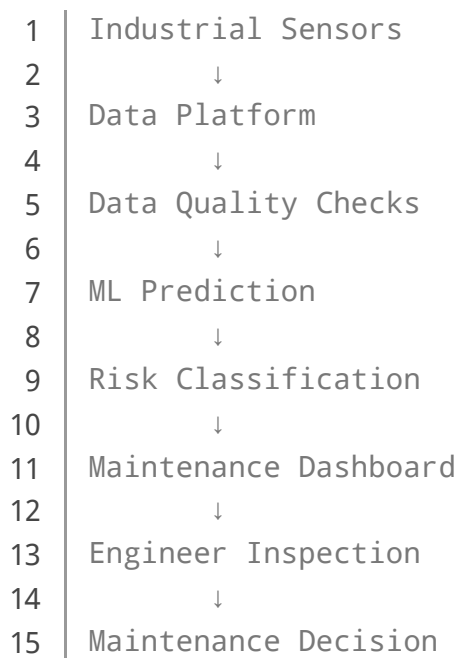
The engineering team collects:

- Vibration measurements
- Temperature readings
- Operating conditions
- Maintenance history
- Failure records
- Production information

The data pipeline cleans and organizes these observations.

Machine-learning models are then trained to identify patterns associated with equipment degradation.

The operational workflow



Why this approach is valuable

The system does not need to replace maintenance engineers.

Instead, it provides an additional analytical layer.

Engineers remain responsible for inspecting the equipment, evaluating safety, and deciding what action should be taken.

This human-in-the-loop architecture is often more practical than attempting complete automation.

What can be learned

The case illustrates an important principle:

AI produces the greatest value when it becomes part of an engineering workflow rather than existing as an isolated model.

Essential Tips

Start with the decision

Ask:

What decision will this AI system improve?

Do not start with:

Which AI model should we use?

Build a baseline

Develop a simple analytical solution before introducing complex deep-learning architectures.

This gives engineers a reference point for measuring improvement.

Treat data as an engineering asset

Create processes for:

- Data validation
- Version control
- Documentation
- Quality monitoring

- Access management

Design for monitoring

A deployed model should have observable performance.

Monitor both the model and the data feeding it.

Keep humans involved where appropriate

AI should support engineering expertise, particularly when decisions involve safety, financial risk, regulatory compliance, or significant operational consequences.

Focus on measurable value

An AI project should have meaningful success criteria such as:

- Reduced downtime
- Better forecasting
- Faster analysis
- Reduced operational cost
- Improved quality
- Faster engineering decisions

The current industry direction is increasingly shifting from AI experimentation toward measurable organizational outcomes and redesigned workflows.

FAQs

What is the role of [machine learning in data analytics](#)?

Machine learning enables analytics systems to identify patterns in historical data and use those patterns for prediction, classification, anomaly detection, recommendation, and other analytical tasks.

Is deep learning always better than machine learning?

No. Deep learning is powerful for complex and large-scale datasets, but traditional machine-learning algorithms can be more efficient, easier to interpret, and highly effective for structured datasets.

What is the biggest AI trend in data analytics?

AI agents, natural-language analytics, multimodal systems, semantic technologies, predictive analytics, and automated decision support are among the major trends shaping modern data and analytics. Garner identifies AI agents, semantics, and converged data-and-analytics platforms as key 2026 trends.

Will AI replace data analysts?

AI is more likely to change the responsibilities of analysts than eliminate the entire profession. Analysts increasingly need to combine data expertise with AI evaluation, business understanding, communication, and critical thinking.

Why is data quality important for AI?

AI models learn from data. If the training data contains systematic errors, missing information, incorrect labels, or biased observations, the resulting system can produce unreliable outputs.

What is multimodal AI analytics?

Multimodal AI analytics combines different data types—such as text, images, audio, video, sensor information, and tables—to create a more comprehensive analytical view.

What is responsible AI?

Responsible AI refers to designing and operating AI systems with appropriate attention to reliability, transparency, fairness, privacy, security, accountability, and governance.

Should companies use AI agents for autonomous decisions?

Only when the decision is appropriate for automation and sufficient safeguards exist. High-impact decisions generally require stronger controls, monitoring, auditability, and human oversight.

Conclusion

Artificial intelligence is fundamentally changing data analytics.

Machine learning provides powerful capabilities for prediction, classification, anomaly detection, and forecasting. Deep learning expands analytical capabilities for complex datasets such as images, signals, speech, and large-scale time-series information.

Generative AI and AI agents are adding new interaction models in which users can communicate with analytical systems using natural language.

However, the future of AI analytics is not simply about using bigger models.

The strongest engineering systems will combine:

High-quality data + appropriate ML/DL models + reliable infrastructure + domain expertise + governance + continuous monitoring.

The emerging AI-first enterprise is therefore better understood as an integrated engineering ecosystem rather than a single algorithm. Garner's 2026 data-and-analytics outlook points toward organizations increasingly embedding AI into decisions, workflows, semantics, and data platforms.

For students and professionals, this creates an important opportunity. Learning machine learning and deep learning remains valuable, but modern data professionals also need to understand data engineering, model evaluation, AI governance, deployment, security, and business or engineering context.

The most successful AI analytics systems will not merely produce more predictions.

They will help people make better decisions—faster, more reliably, and with appropriate human control. 🚀🤖📊

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